

How to write a report

Linda McPhee

Linda Silvertand

In general

- Writing and editing are two separate processes
- Writing and editing at the same time? → writers block
- Start with a draft
 - A draft is not perfect, do not try to make it perfect
 - Content: general topics of the story you want to tell
 - Draft 1: YOU need to understand the first draft; write it accordingly
 - Put the draft away for a couple of days
 - Draft 2 = editing → look at your paper through the eyes of someone else. Does this 'other person' understand what is written?

Draft - Exercise

The main goal(s) of your project have been formulated.

Individually:

- Make a sketch of how your poster would look like with the most ideal results/experiments to show that you have accomplished the these goals.
- Base your poster on the knowledge you now have attained and ignore the results you have gained in the past weeks.
- Include also the experiments you still need to do to accomplish you final goal(s)

Within group/duo:

- Discuss the differences and similarities between the ideas.
- Try to reach a consensus on what to put in this poster?
- Does this change the course of experiments you have planned?

Draft - exercise

Individually:

- Make a draft of a poster of the **results for your project so far**.
- What would you present? Which data are relevant to reach your goals?

Group:

- Compare and discuss the posters
- What data are missing, what data need confirmation, which experiments are the most informative?

Drafts

- Draft 1: write the piece so YOU understand it
- Draft 2: Edit through the eyes of someone else. Be your reader.
 - Language (grammar, style, etc)
 - More elaboration on topics that seem obvious to you?
- Draft 3: Let someone else read it
 - Do they understand what is being said?
 - Structure and contents: logical structure, missing topics, etc?
 - Let someone else read it out loud: linguistic obstacles (incorrect grammar, sentences too long or complex, etc)

Subjects

- Introduction
- M&M
- Results
- Discussion
- Conclusion
- references
- Summary

Introduction

Answer the following questions:

1. **Where are we?** (what is the situation/context; try to make a connection to the background of your reader (the teacher that will be grading your paper))
2. Anamnesis
 - a) What has been done already and what do we know: → literaturesurvey (ca 15-20 papers of reviews, what are the usual strategies, necessary background to understand rest of the paper, etc)
 - b) Where is the discrepancy in 2a and how do we fill this discrepancy with our project? Describe the discrepancy in other papers/research.
3. Yer own research
 - a) Focus the attention of the reader to your own research by describing how your research fills this discrepancy.
 - b) Reveal the most important details of your own research (important conclusions, development of a new method or application or the read thread of your experiments). This encourages the reader to read on.

Where are we- Exercise

- Group:
 - Zoom out: Where in the bigger picture of this topic are we; describe the bigger picture and where in the whole process are we?
 - What is the situation: a clear and SMART formulation of your goal (what do you want to achieve with your research).
 - How to best explain your reader what you have done?
 - Who is your reader?
 - What is the background of your reader? What additional information does your reader require to understand this paper and what information can you skip?
 - If you were an outsider to this research, what would you want to know?

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Anamnesis- Exercise

- Group:
 - Make an inventory:
 - What are the topics on your subject?
 - Have you found literature to substantiate your claims in the introduction?
 - Reviews and meta-analysis papers: Most information dense and cover broad topics → most useful to introduction
 - Peer reviewed literature: individual research
 - Other sources: can be used, but make sure they are verified by experts in the field (university websites, SOPs for industry, application notes, etc)
 - What information is useful to your subject and what not?
 - Write a summary of this relevant background information (homework)
 - Where are the shortcomings/discrepancies in research (see also your project description)
 - This summary is the introduction to your sales pitch, because:
 - This summary shows the shortcomings of other research and you now can easily create your own unique selling point: what in your research is done to solve the shortcomings/discrepancies of other research?

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- 3. Your own research**
 - a) Focus the attention of the reader to your own research by **describing how your research fills this discrepancy.**
 - b) **Reveal the most important details of your own research** (important conclusions, development of a new method or application or the read thread of your experiments).

This encourages the reader to read on.

Own research- exercise

- Group:
 - Discuss: how does our research fill the discrepancy?
- What details do you want to share:
 - Save the best for last? High chance of reader stopping before the best part of the paper.
 - Reveal the best data? Get the reader curious about how you went about to get these fantastic results
 - Content: best results, new insights, new application, systematic proces, etc

Introduction

- Get the 5 answers on paper and mould this into a STORY
- A story:
 - Is **coherent**; the topics are all related, the use of language and layout is the same throughout the introduction, etc
 - Has a certain **flow**; the order of the topics is logical and directed towards the revelation of your data and conclusions. The topics flow into each other smoothly. (introduction: 1-2-3)
 - **Readable**: the language should not be a barrier in understanding of the content. Avoid complex use of language and long sentences. Check for grammar, style, etc (read, read, read and then read again).
 - **Clear**: leave nothing open for the interpretation of the reader. Not everyone can read between the lines or interprets subjects the same way you do.
 - **Is adjusted to the reader**. Know your reader.

Introduction

Writing the introduction is easy, editing takes the most time and effort since there is a lot of reading and evaluating involved.

Start at least 3-4 weeks before handing in the paper with the introduction.

It saves you unnecessary experiments, gets you a clear focus, but mainly saves you the trouble of a required second version.

DO NOT write the paper in one go, briefly check for any major inconsistencies and then hand it in.

Materials and Method

- This couples the reason for the paper (introduction) to the proof (results)
- This is where you make your story believable by:
 - Formulating systematically all the experiments that were performed in order for;
 - Someone else to be able to repeat these experiments and get the same results
- Questions to answer:
 - What have you done to acquire the data

M&M - Exercise

- Group:
 - What data does someone else need to copy my experiments and get the exact same results?
 - Make a list of this information
 - How can you write this information down in the shortest way possible?
 - Examples: look in literature

Keep in mind that a trained professional will be repeating the experiments in a lab in a different country.

Examples M&M

2 Materials and methods

2.1 Materials

Pharmalyte[™] (36% w/v, pH 3–10, copolymer of glycine, glycyglycine, various amines and epichlorohydrin) was purchased from Amersham Bioscience (Piscataway, NJ, USA). Tween 20 (for synthesis) was obtained from Merck KGaA (Darmstadt, Germany). Hydroxyethyl cellulose (HEC, medium viscosity), phosphoric acid (85%), TFA (>99%), sinapic acid (>99%), TEMED (~99%), sodium hydroxide and all proteins (see Table 1) were purchased from Sigma-Aldrich. GlucaGen (Novo Nordisk) was kindly donated by Drs L. J. F. Silvertand and Drs F. N. Sikke (Apotheek Born, Born, The Netherlands). HPLC gradient grade ACN and methanol (MeOH) were purchased from Biosolve (Valkenswaard, The Netherlands). High-purity water (Ultrapure, >18.2 MΩcm) was obtained from a Millipore Synergy UV water purification system (Millipore, Bedford, MA, USA).

2.2 Apparatus and methods

cIEF separations were performed with a PrinCE 500 dual-lift CE system using Dax Data Acquisition and Analysis Software version 6.0 (Prince Technologies, Emmen, The Netherlands), equipped with a Knauer K-2501 UV detector set at 280 nm (Knauer GmbH, Berlin, Germany) and an eCAP neutral capillary (50 μm id, 40.2 cm total length, Beckman Coulter, Fullerton, CA, USA). Prior to each run the capillary was rinsed first with 10 mM phosphoric acid in 0.3% HEC, then with purified water and finally with 0.1 % Tween 20 in 0.3% HEC. Subsequently, the capillary was filled with Sample 1 (ribonuclease (RNase, 90 μg/mL), myoglobin (Myo), carbonic anhydrase II (CAII), β-lactoglobulin B (LGB) and α-lactalbumin (Lac) (all at 10 μg/mL), Pharmalyte[™] (1% w/v), TEMED (0.1% v/v), HEC (0.3% w/v) and Tween 20 (0.1% w/v)) or with Sample 2 (= Sample 1, but LGB and Lac are replaced with β-lactoglobulin A (LGA) (16 μg/mL) and cholecystokinin flanking peptide (CCK) (8 μg/mL)). The inlet vial (anolyte) was filled with 10 mM phosphoric acid in 0.3% HEC. The capillary outlet was inserted in the hollow needle (electrically grounded) of the CE interface of an LC Packings Probot Microfraction Collector (Dionex, Amsterdam, The Netherlands) (see also Fig. 1). Sheath liquid (200 mM ammonium hydroxide in 50% MeOH and 0.01% Tween 20) was delivered through this needle *via* the syringe of the Probot Spotter and was set to deposit 0.5 μL sheath liquid *per* spot throughout the entire run. During focusing the capillary outlet and needle were submerged in the outlet vial (catholyte: 200 mM ammonium hydroxide in 50% MeOH). Focusing was performed at 20 kV at room temperature for 15–90 min. After focusing, the catholyte waste vial was removed and the separated proteins were mobilized at 20 V and 100 mbar air pressure towards an ABI stainless steel MALDI target plate (Applied Biosystems, Framingham, MA, USA) with 192+6 predefined spots. Upon completion of the spotting procedure, 0.75 μL MALDI matrix (5 mg/mL sinapic acid in 80% v/v ACN with 0.1% v/v TFA) was added *per* spot and dried.

MALDI-TOF MS experiments were performed with an Applied Biosystems 4700 Proteomics Analyzer. The *m/z* recording range was set to 5000–40 000 with a focus *m/z* of 20 000 in linear mode. MALDI-TOF spectra were analyzed using Data Explorer software Version 3.0 (Applied Biosystems).

Examples M&M

Chemicals

Cyclophosphamide monohydrate reference standard and d4-cyclophosphamide (Fig. 1(D)), internal standard for cyclophosphamide, originated from Baxter Oncology GmbH (Frankfurt am Main, Germany). Fludarabine reference standard was obtained from Sigma-Aldrich Chemie GmbH (Steinheim, Germany). The Leustatin[®] formulation (cladribine, Fig. 1(B)) was used as the internal standard for fludarabine and was purchased from Janssen-Cilag BV (Tilburg, The Netherlands). Acetonitrile and methanol (both supra-gradient grade) were from Biosolve Ltd. (Amsterdam, The Netherlands). Ammonium acetate, 25% ammonia, 100% acetic acid, 37% hydrochloric acid (all analytical grade) and LiChrosolve water (LC gradient grade) originated from Merck (Darmstadt, Germany). Drug-free human heparin control plasma and drug-free human EDTA control plasma originated from Sanquin (Amsterdam, The Netherlands).

Instrumentation

The LC system consisted of a model 1100 series binary LC pump, an 1100 series cooled autosampler (10°C) and an 1100 series mobile phase degasser (all Agilent Technologies, Palo Alto, CA, USA). A stepwise gradient was applied to elute the analytes. At time zero a mixture of 96% 1 mM ammonia solution and 4% acetonitrile was flushed through the column. After 2 min, the percentage of acetonitrile was increased to 25% in 1 min and maintained at this percentage for 3 min. From 6 to 6.1 min, the percentage of acetonitrile was set back to 4% and was pumped through the column during the last 6 min of the run. The flow was set at 400 µL/min. The analytes were separated on an Extend C18 column (150 × 2.1 mm i.d., 5 µm particle size), with an Extend C18 pre-column (12.5 × 2.1 mm i.d., 5 µm particle size) (Agilent Technologies) protected by a 5 µm in-line filter (Upchurch Scientific, Oak Harbor, WA, USA). The column was connected to the ESI source of a API 3000 triple quadrupole mass spectrometer (PE Sciex, Thornhill, ON, Canada) through a divert valve (Valco Instruments, Houston, TX, USA). The valve was used to prevent the introduction of endogenous compounds into the mass spectrometer by directing the eluent flow the first 2 min and the last minute to waste. The injection volume was 10 µL. Multiple reaction monitoring (MRM) chromatograms were used for quantification using Analyst[™] software version 1.2 (PE Sciex). Standard solutions of the compounds were infused directly into the ESI source to determine positively charged precursor ions. Only singly charged ions were observed. Product ions were determined by subjecting the precursor ions to collision-induced dissociation. Transitions of the precursor-product ion pairs are listed in Table 1 and ESI-MS/MS settings are listed in Table 2.

Preparation of stock solutions, working solutions, calibration standards and quality control samples

Stock solutions of cyclophosphamide were prepared independently by dissolving cyclophosphamide monohydrate in methanol at a concentration of 1.0 mg/mL. One solution was used to spike plasma calibration standards and one was prepared independently to spike quality control (QC) samples. The stock solutions for fludarabine were prepared in acidified methanol (10 mM HCl in methanol).

Working solutions were prepared from stock solutions and contained both cyclophosphamide and fludarabine at concentration levels of 300, 2500 and 7500 ng/mL in methanol for the preparation of QC samples and 100, 200, 500, 1000, 5000 and 10 000 ng/mL for the preparation of calibration standards. All solutions were stored at −20°C. Plasma QC samples and calibration standards were prepared by adding working solutions to control human heparin plasma or control human EDTA plasma. QC samples were prepared at low, medium and high concentration levels (3, 25 and 75 ng/mL) and additional levels around the lower limit of quantitation (LLQ, 1 ng/mL) and at a concentration equal to 25 times the upper limit of quantitation (ULQ, 100 ng/mL). Calibration working solutions were used to spike human control plasma to obtain concentration levels for calibration standards of 1, 2, 5, 10, 50, 75 and 100 ng/mL.

A stock solution of d4-cyclophosphamide was prepared by dissolving d4-cyclophosphamide in methanol at a concentration of 1 mg/mL. A working solution of d4-cyclophosphamide was prepared by diluting the stock solution with methanol to a concentration of 40 µg/mL. Cladribine was already in solution at a concentration of 1 mg/mL (Leustatin[®]). This stock solution was diluted with methanol to obtain a cladribine working solution of 20 µg/mL. The final working solution was obtained by adding 100 µL 40 µg/mL d4-cyclophosphamide working solution to 100 µL 20 µg/mL cladribine working solution and adding methanol up to 10.0 mL. Plasma standards and samples and stock solutions of fludarabine, cladribine, cyclophosphamide and d4-cyclophosphamide were stored at −20°C.

Table 2. MS/MS settings

Parameter	Setting				
Duration	12 min				
Ionspray voltage (positive ion mode)	+5 kV				
Curtain gas (N ₂)	1.2 L/min				
Temperature	400°C				
Nebulizer gas (compressed air)	1.8 L/min				
Turbo gas (N ₂)	7.0 L/min				
Collision-activated dissociation gas (N ₂)	8 psi				
Channel electron multiplier	1900 V				
Parameter	Fludarabine	Cladribine	Cyclophosphamide	d4-Cyclophosphamide	
Dwell time (ms)	50	50	50	50	
Declustering potential (V)	36	31	41	56	
Focusing potential (V)	160	150	180	240	
Entrance potential (V)	10	10	10	10	
Collision energy (V)	25	21	31	31	
Collision cell exit potential (V)	10	12	36	26	

Results

- Combined with discussion?

R&D&C&R - topics

- What did I find?
- Is it surprising?
- Similarities and differences between the [different experiments](#). Inconsistencies?
- Similarities and differences compared to [different groups that performed the same experiments](#). Inconsistencies?
- Similarities and differences compared to [peer-reviewed literature on the same topic](#). Inconsistencies?
- Explain the differences and similarities and focus on the reliability of the data:
 - Are the methods reliable?
 - What data are most reliable
 - Try and find an explanation for EVERYTHING you found (also for the 'unusable' data). Can you already reduce the list of explanations by comparing data from different methods or experiments? Is there one explanation that seems to stand out after deduction?
- Generalise
- Implications
- Next steps

R&D&C&R - topics

- What did I find (RESULTS)?
- Is it surprising?
- Similarities and differences between the different experiments. Inconsistencies?
- Similarities and differences compared to different groups that performed the same experiments. Inconsistencies?
- Similarities and differences compared to peer-reviewed literature on the same topic. Inconsistencies?
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Results – Exercise 1

- Select one result from your labjournal.
- Write a description of this result in text.
- Exchange descriptions and try to recreate the result from this description.
- Evaluate: is all the relevant information depicted in the reconstructed image?

Results – Exercise 2

- Select one result from your labjournal.
- Write your main conclusion down
- Exchange the result with another student.
- Write down what your conclusion is based on the result from someone else only.
- Write down what information you need to correctly interpret the new result
- Evaluate with the other student:
 - Make a list of basic information required to interpret the result → this will be the minimal information required for the caption of the result
 - Discuss the differences in interpretation: why is there a difference, what does the other student need to know so that they understand YOUR interpretation?
 - Define a strategy for yourself on how to steer the reader towards your interpretation

Discussion

- What did I find (results)?
- Is it surprising → ① **your own opinion on the results?**
 - Similarities and differences between the different experiments. Inconsistencies?
- What have the others found? ② **peer-opinion on the results?**
 - Similarities and differences compared to different groups that performed the same experiments. Inconsistencies?
- What do the experts say? → ③ **expert-opinion on the results?**
 - Similarities and differences compared to **peer-reviewed literature** on the same topic. Inconsistencies?
- Explain the differences and similarities and focus on the reliability of the data:
 - Try and find an **explanation** for EVERYTHING you found (also for the 'unusable' data and don't forget the 'obvious interpretations'). Can you already reduce the list of explanations by comparing data from different methods or experiments? Is there one explanation that seems to stand out after deduction?
 - Explain **inconsistencies**
 - **Reliability:**
 - Are the methods reliable?
 - What are the limitations in your work?
 - What data are most reliable
- Generalise
- Implications
- Next steps

Exercise discussion

- Group: select one of the results that is difficult to interpret.
- Own opinion:
 - Individual: take 5 minutes to think of what this result means: write down how you'd interpret these data. What are the limitations and implications ?
 - What did you expect before the experiment was performed, are the results similar or different
 - Discuss your findings in the group.
- Peer opinion
 - Comparing these results to other results, are they similar or different?
 - Do the results fit the results obtained with other techniques/ experiments?
 - What are the limitations/reliability of the other experiments?
 - What if you compare these data with data obtained by other groups?
- Expert opinion
 - What kind of information would you need to verify (substantiate or discard) your results?
 - How can you find this information: what sources would you use in your search and write down a couple of keywords to search for.
 - Can you think of alternative searching strategies if there is nothing to find in this specific area?

Conclusion

- What did I find (results)?
- Discussion:
 - your **own** opinion on the results?
 - **peer**-opinion on the results?
 - **expert**-opinion on the results?
- Explain the differences and similarities
- Generalise (conclusion)
 - What conclusion can be drawn from ALL sources? Mention also the reliability of these data
- Implications (global) (conclusion)
 - What global implications does your research have?
- Next step → Recommendations
 - What will be the global logical next steps in this research?
 - How would you go about executing these next steps (more focused on practical planning)
 - How would you solve your own limitations?

Exercise conclusion

- A conclusion is **NOT** a summary of your results and discussion
- Assemble all data and formulate a weighted interpretation. Reliability of your data is an important factor of your conclusion. Try and find the most plausible explanation for your results, preferably underlined with peer-reviewed literature.
- Implications: If your findings were absolutely true, how would this impact the rest of the world. Think globally and big for example: would your new technique be able to see a certain diseasemechanisme, would it be a faster or more sensitive analysis, would it change the way you'd formulate drugs, etc.

Recommendations

- What did I find (results)?
- Discussion:
 - your **own** opinion on the results?
 - **peer**-opinion on the results?
 - **expert**-opinion on the results?
- Explain the differences and similarities
- Generalise (conclusion)
- Implications (global) (conclusion)
- Next step → Recommendations
 - What will be the global logical next steps in this research?
 - How would you go about executing these next steps (more focused on practical planning)
 - How would you solve your own limitations?

Recommendations - exercises

- Discuss within the group
 - What would be the next step in this project? (cell-line experiments, other data, formulation experiments, validation, etc?)
 - How would you plan and perform this?
 - What are the shortcomings in your own research and how would you solve these? What data do you need to complete the entire story, what does the next person in the line of this project need to do? To answer these questions, try to look ahead in the chain of the project.

Summary

- Attract the attention of the reader in a haystack of other papers
- Be short, compact and concise
- Content (4-5 sentences):
 1. What have you been trying to do (or why did you want to do this) (goal)
 2. What DID you actually do (method)
 3. What did you find (results)
 4. What does this mean (discussion/conclusion)

Summary - example

GOAL

Fludarabine and cyclophosphamide are anticancer agents mainly used in the treatment of hematologic malignancies. We have developed and validated an assay using high-performance liquid chromatography (HPLC) coupled with electrospray ionization tandem mass spectrometry for the quantification of fludarabine in combination with cyclophosphamide in human heparin and human EDTA plasma. Sample pre-treatment consisted of a protein precipitation with cold acetonitrile (-20°C) using 250 μL of plasma. Separation was performed on an Extend C18 column (150×2.1 mm i.d.; 5 μm) with a stepwise gradient using 1 mM ammonia solution and acetonitrile at a flow rate of 400 $\mu\text{L}/\text{min}$. The analytical run time was 12 min. The triple quadrupole mass spectrometer was operated in the positive ion mode and multiple reaction monitoring was used for drug quantification. The method was validated over a concentration range of 1 to 100 ng/mL for fludarabine and cyclophosphamide in human heparin and human EDTA plasma. The coefficients of variation were $<13.9\%$ for inter- and intra-day precisions. Mean accuracies were also within the designated limits ($\pm 15\%$). The analytes were stable in plasma, processed extracts and in stock solution under all relevant conditions. Copyright © 2005 John Wiley & Sons, Ltd.

Summary - example

METHOD

Fludarabine and cyclophosphamide are anticancer agents mainly used in the treatment of hematologic malignancies. We have developed and validated an assay using high-performance liquid chromatography (HPLC) coupled with electrospray ionization tandem mass spectrometry for the quantification of fludarabine in combination with cyclophosphamide in human heparin and human EDTA plasma. Sample pre-treatment consisted of a protein precipitation with cold acetonitrile (-20°C) using 250 μL of plasma. Separation was performed on an Extend C18 column (150×2.1 mm i.d.; 5 μm) with a stepwise gradient using 1 mM ammonia solution and acetonitrile at a flow rate of 400 $\mu\text{L}/\text{min}$. The analytical run time was 12 min. The triple quadrupole mass spectrometer was operated in the positive ion mode and multiple reaction monitoring was used for drug quantification. The method was validated over a concentration range of 1 to 100 ng/mL for fludarabine and cyclophosphamide in human heparin and human EDTA plasma. The coefficients of variation were $<13.9\%$ for inter- and intra-day precisions. Mean accuracies were also within the designated limits ($\pm 15\%$). The analytes were stable in plasma, processed extracts and in stock solution under all relevant conditions. Copyright © 2005 John Wiley & Sons, Ltd.

Summary - example

RESULTS

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Summary - example

CONCLUSION

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Development and characterization of cIEFMALDI- TOF MS for protein analysis

This paper describes the hyphenation of cIEF and MALDI-TOF MS via a fractionation or spotting device. After focusing in cIEF the compounds are hydrodynamically mobilized and deposited on a MALDI target plate using a sheath liquid interface, which provides the catholyte solution and the electrical ground. From previous experiments, sample conditions that resulted in a high resolution in cIEF and acceptable protein signal intensity in MS were selected [Silverstrand et al., Electrophoresis, 2008, 29, 1985–1996]. Besides the mixture of test proteins, the sample solution contains 1% Pharmalyte, 0.3% hydroxyethyl cellulose and 0.1% Tween 20 and is used for both optimization as well as characterization of the cIEF-MALDI-TOF MS system. Hyphenation problems encountered are mainly due to transfer of the liquid from the needle to the MALDI target plate and are solved by choosing the proper sheath catholyte (200mM NH_4OH in 50% methanol with 0.1% Tween20). MS electropherograms were reconstructed by plotting the intensities of the m/z values corresponding to the proteins versus migration time (related to spot number). Reproducibility, peak width and signal intensity for different focusing and spotting (fractionation) times were calculated using these reconstructed MS electropherograms as well as the UV electropherograms. The best results were obtained with focusing time of 75 min (no under- or overfocusing) and a spotting time of 5 s (highest protein signal intensity in MS). The applicability of the system is demonstrated by the analysis of a biopharmaceutical (glucagon) and its deamidation product.